

Pearson Edexcel International GCSE

Mathematics A

Modular Unit 1

June 2022 Reconstructed Past Paper

DETAILED SOLUTIONS

Original question followed by a full worked solution

29 adaptive question sections 100 marks

Selected from Linear Higher Papers 1H and 2H

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L-to-M | CONVERTED MODULAR EDITION

Selected from Linear Higher papers; not an original Modular examination paper.

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About this reconstructed paper

Each original source section is followed by the demand-local candidate solution and final answer in the same order as the Student book.

Ownership rule: Unit 2 owns a demand when its assessed or terminal statement is Unit 2, even if Unit 1 skills are prerequisites. This book contains the demands owned by Unit 1; the selection contains 100 marks.

This is an Elite reconstructed teaching paper selected from official Linear Higher material; it is not an official Pearson Modular examination paper. Candidate source assets are retained for private teaching use only.

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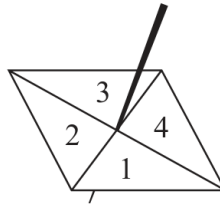
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Original question 01

4MA1-1H Q2 Q2

ORIGINAL QUESTION**4 marks**

- 2 Here is a biased 4-sided spinner.



The table gives the probabilities that, when the spinner is spun once, it will land on 1 or it will land on 3

Number	1	2	3	4
Probability	0.26		0.18	

The probability that the spinner will land on 2 is equal to the probability that the spinner will land on 4

Ravina is going to spin the spinner a number of times.

Ravina works out that an estimate for the number of times the spinner will land on 3 is 45

Work out an estimate for the number of times the spinner will land on 4

Worked solution 01

Chapter: Probability Toolkit

MODULAR UNIT 1

4 marks

Q2 – Chapter: Probability Toolkit

Family: Use expected frequency • Variant: Reverse expected frequency to find trials or probability

MethodLet the probability of landing on 2, and also on 4, be p .*Why: The question states that these two probabilities are equal.***Method** $2p + 0.26 + 0.18 = 1$, so $2p = 0.56$ and $p = 0.28$.*Why: All four spinner probabilities add to 1.***Method**If 45 is the estimated frequency for outcome 3, the estimated number of spins is $45/0.18 = 250$.*Why: Estimated frequency equals probability multiplied by the number of trials.***Method**The estimated frequency for outcome 4 is $250 \times 0.28 = 70$.*Why: Apply the probability of outcome 4 to the same number of spins.***Final answer****estimated frequency: 70**

Original question 02

4MA1-1H Q5 Q5(a)

ORIGINAL QUESTION**2 marks**

5 (a) Expand and simplify $(n - 6)(n + 4)$

.....
(2)

Worked solution 02

Chapter: Expanding Brackets

MODULAR UNIT 1

2 marks

Q5(a) – Chapter: Expanding Brackets

Family: Expand two linear expressions • Variant: Standard double bracket

Method

$$(n - 6)(n + 4) = n^2 + 4n - 6n - 24.$$

*Why: Distribute each term in the first bracket across the second.***Method**Collecting like terms gives $n^2 - 2n - 24$.*Why: Combine $4n$ and $-6n$.***Final answer**expansion: $n^2 - 2n - 24$

Original question 03

4MA1-1H Q5 Q5(b)

ORIGINAL QUESTION**3 marks**

(b) Solve $2x - 3 = \frac{3x - 5}{4}$

Show clear algebraic working.

$x =$
(3)

Worked solution 03

Chapter: Solving Linear Equations

MODULAR UNIT 1

3 marks

Q5(b) – Chapter: Solving Linear Equations

Family: Balance a linear equation • Variant: Linear unknown on both sides

MethodFrom $2x - 3 = (3x - 5)/4$, multiply by 4 to get $8x - 12 = 3x - 5$.*Why: Clearing the denominator preserves the equality.***Method** $8x - 3x = 12 - 5$, so $5x = 7$.*Why: Collect the x-terms on one side and constants on the other.***Method**Therefore $x = 7/5$.*Why: Divide both sides by 5.***Final answer****x:** $x = 7/5$

Original question 04

4MA1-1H Q9 Q9(a)

ORIGINAL QUESTION**1 marks**

9 (a) Simplify $8 \times (4t)^0$

(1)

Worked solution 04

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

1 marks

Q9(a) – Chapter: Algebraic Roots & Indices

Family: Interpret zero, negative and fractional indices • Variant: Zero and negative indices

MethodSince $(4t)^0 = 1$, $8 \times (4t)^0 = 8$.*Why: Any nonzero base raised to the power zero equals 1.***Final answer**

simplified value: 8

Original question 05

4MA1-1H Q9 Q9(b)

ORIGINAL QUESTION**1 marks**

$$x^6 \div x^{-5} = x^p$$

(b) Find the value of p $p = \dots\dots\dots$
(1)

Worked solution 05

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

1 marks

Q9(b) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Multiply or divide common bases

Method

$$x^6 / x^{-5} = x^{(6 - (-5))} = x^{11}, \text{ so } p = 11.$$

*Why: When dividing like bases, subtract the denominator exponent.***Final answer**

p: $p = 11$

Original question 06

4MA1-1H Q9 Q9(c)

ORIGINAL QUESTION**2 marks**(c) Simplify fully $(2k^2m^4)^3$

(2)

Worked solution 06

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

2 marks

Q9(c) – Chapter: Algebraic Roots & Indices

Family: Combine indexed algebraic factors • Variant: Power of a product or power

Method

$$(2k^2 m^4)^3 = 2^3 (k^2)^3 (m^4)^3.$$

*Why: Apply the outside power to every factor.***Method**This simplifies to $8k^6m^{12}$.*Why: Multiply the nested exponents and evaluate 2^3 .***Final answer**simplified expression: $8k^6m^{12}$

Original question 07

4MA1-1H Q11 Q11(a)

ORIGINAL QUESTION**2 marks****11** (a) Factorise $9x^2 - 4y^2$

(2)

Worked solution 07

Chapter: Factorising

MODULAR UNIT 1

2 marks

Q11(a) – Chapter: Factorising

Family: Factorise a non-monic or special quadratic • Variant: Difference of two squares

Method

$$9x^2 - 4y^2 = (3x)^2 - (2y)^2.$$

*Why: Recognise a difference of two squares.***Method**

$$\text{Therefore } 9x^2 - 4y^2 = (3x + 2y)(3x - 2y).$$

*Why: Use $a^2 - b^2 = (a + b)(a - b)$.***Final answer****factorisation:** $(3x + 2y)(3x - 2y)$

Original question 08

4MA1-1H Q11 Q11(b)

ORIGINAL QUESTION**3 marks**

(b) Express $\frac{7}{8} - \frac{x+3}{4x}$ as a single fraction in its simplest form.

.....
(3)

Worked solution 08

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

Q11(b) – Chapter: Algebraic Fractions

Family: Add or subtract algebraic fractions • Variant: Linear denominators

MethodUse denominator $8x$: $\frac{7}{8} - \frac{(x + 3)}{(4x)} = \frac{7x}{(8x)} - \frac{2(x + 3)}{(8x)}$.*Why: $8x$ is a common denominator for 8 and $4x$.***Method**Combining gives $\frac{[7x - 2(x + 3)]}{(8x)} = \frac{(7x - 2x - 6)}{(8x)}$.*Why: Keep the subtraction outside the bracket when combining numerators.***Method**The single simplified fraction is $\frac{(5x - 6)}{(8x)}$.*Why: Collect the numerator terms.***Final answer****single fraction:** $\frac{(5x - 6)}{(8x)}$

Original question 09

4MA1-1H Q12 Q12(a), Q12(b)

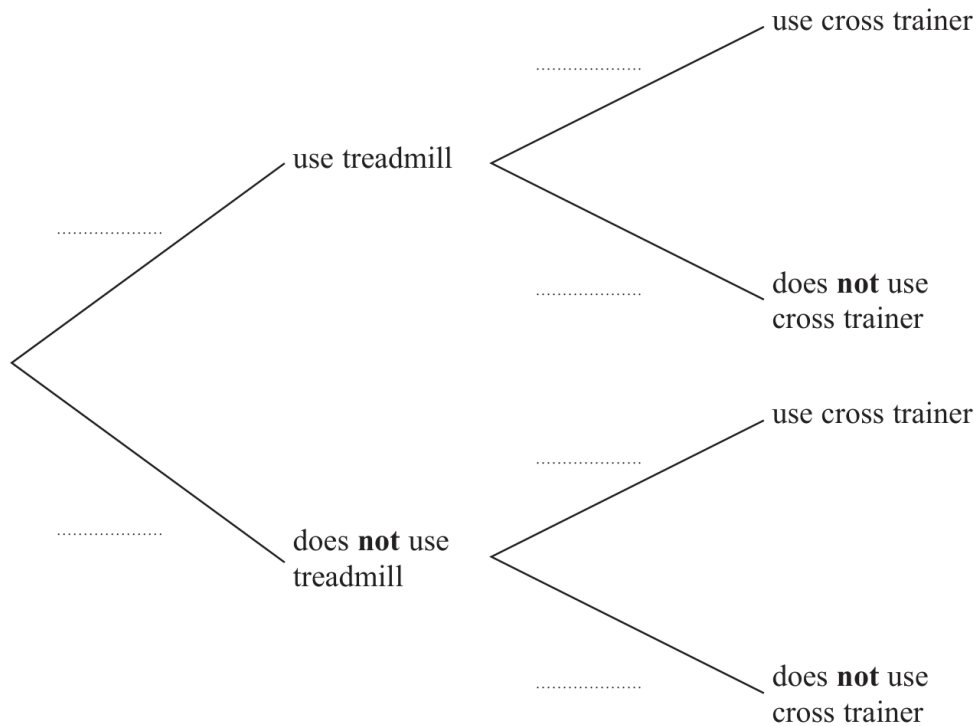
ORIGINAL QUESTION**4 marks****12** Rudolf goes to the gym.

The probability that he will use the treadmill is 0.8

When he uses the treadmill, the probability that he will use the cross trainer is 0.3

When he does **not** use the treadmill, the probability that he will use the cross trainer is 0.6

(a) Complete the probability tree diagram for this information.

**(2)**

(b) Work out the probability that Rudolf uses both the treadmill and the cross trainer.

(2)

Worked solution 09

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q12(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Construct and read probability trees • Variant: Construct a tree from conditional wording

Method

The first split is treadmill 0.8 and no treadmill 0.2.

*Why: Complementary probabilities on a split add to 1.***Method**

After treadmill, use cross trainer 0.3 and not 0.7; after no treadmill, use cross trainer 0.6 and not 0.4.

*Why: Complete each pair with its complement.***Final answer****tree probabilities:** 0.8/0.2; then 0.3/0.7 and 0.6/0.4**Q12(b) – Chapter: Combined & Conditional Probability**

Family: Use conditional probability and restricted sample spaces • Variant: Use branch-dependent conditional probabilities

MethodThe path for both activities is treadmill followed by cross trainer, with probability 0.8×0.3 .*Why: Multiply probabilities along one complete path.***Method** $0.8 \times 0.3 = 0.24$.*Why: Evaluate the path product.***Final answer****probability:** 0.24

Completed diagram 09

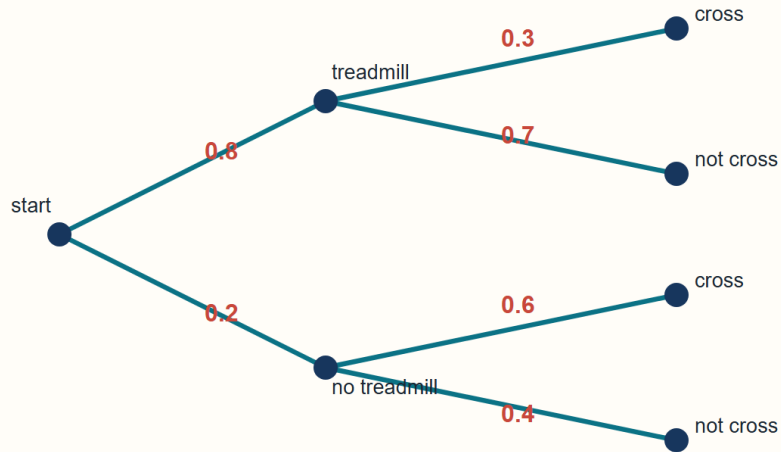
Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q12(a) Completed probability tree

Place each complement beside its branch

Each pair is complementary: $0.8+0.2=1$, $0.3+0.7=1$, $0.6+0.4=1$.

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Original question 10

4MA1-1H Q15 Q15

ORIGINAL QUESTION**4 marks**

- 15** Abraham is going to play a computer game.
Abraham can win the game, draw the game or lose the game.

For any game that Abraham plays

the probability that he wins the game is 0.3
the probability that he draws the game is 0.5
the probability that he loses the game is 0.2

When Abraham wins a game, he scores +10 points.

When Abraham draws a game, he scores 0 points.

When Abraham loses a game, he scores −5 points.

Abraham plays 3 games and the points he scores in each of the 3 games are added together to get his total score.

Work out the probability that when he has played 3 games his total score is 0 points.

Worked solution 10

Chapter: Combined & Conditional Probability

MODULAR UNIT 1

4 marks

Q15 – Chapter: Combined & Conditional Probability

Family: Combine independent events • Variant: Find exactly one, same or different independent outcomes

Method

A total of zero occurs for three draws, with probability $0.5^3 = 0.125$.

Why: Three zero-point outcomes add to zero.

Method

It also occurs for one win and two losses because $10 - 5 - 5 = 0$.

Why: Identify the other score combination that totals zero.

Method

There are 3 positions for the win, so this probability is $3 \times 0.3 \times 0.2^2 = 0.036$.

Why: Count all permutations of one win and two losses.

Method

The required probability is $0.125 + 0.036 = 0.161$.

Why: The two cases are mutually exclusive, so add their probabilities.

Final answer

probability: 0.161

Original question 11

4MA1-1H Q16 Q16

ORIGINAL QUESTION**3 marks**

- 16 Without using a calculator, show that $\frac{12}{\sqrt{2}-1} - (\sqrt{2})^5 = 2\sqrt{32} + 12$
Show your working clearly.

Worked solution 11

Chapter: Surds

MODULAR UNIT 1

3 marks

Q16 – Chapter: Surds

Family: Simplify and collect surds • Variant: Add or subtract like surds

Method

$$12/(\sqrt{2} - 1) = 12(\sqrt{2} + 1)/[(\sqrt{2} - 1)(\sqrt{2} + 1)] = 12\sqrt{2} + 12.$$

*Why: Multiply by the conjugate to rationalise the denominator.***Method**

$$(\sqrt{2})^5 = 4\sqrt{2}, \text{ so the left side is } 12\sqrt{2} + 12 - 4\sqrt{2} = 8\sqrt{2} + 12.$$

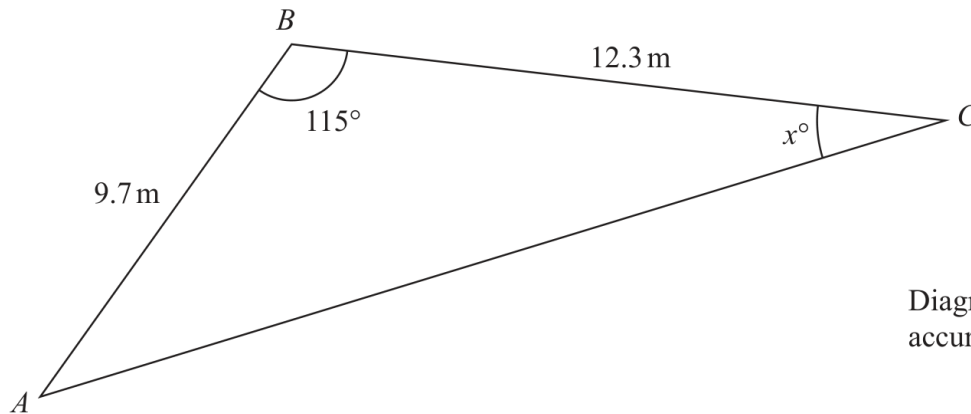
*Why: Use powers of $\sqrt{2}$ and collect like surds.***Method**

Since $\sqrt{32} = 4\sqrt{2}$, the right side $2\sqrt{32} + 12$ is also $8\sqrt{2} + 12$; therefore the identity is shown.

*Why: Simplify both sides to the same exact surd form.***Final answer****shown identity:** Both sides simplify to $8\sqrt{2} + 12$

Original question 12

4MA1-1H Q18 Q18

ORIGINAL QUESTION**5 marks****18** Here is triangle ABC Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to 3 significant figures.

 $x =$

Worked solution 12

Chapter: Sine, Cosine Rule & Area of Triangles

MODULAR UNIT 1

5 marks

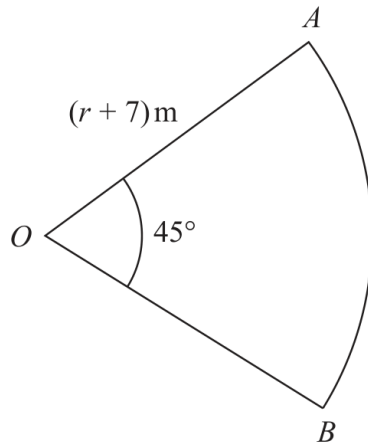
Q18 – Chapter: Sine, Cosine Rule & Area of Triangles

Family: Use the sine rule • Variant: Find an acute angle using the sine rule

MethodBy the cosine rule, $AC^2 = 9.7^2 + 12.3^2 - 2(9.7)(12.3)\cos(115 \text{ degrees})$.*Why: The two known sides enclose the known 115-degree angle.***Method** $AC^2 = 346.2...$, so $AC = 18.606...$ m.*Why: Evaluate the cosine-rule expression without premature rounding.***Method**Using the sine rule, $\sin(x)/9.7 = \sin(115 \text{ degrees})/18.606...$.*Why: Side $AB = 9.7$ is opposite angle x , and AC is opposite 115 degrees.***Method**Therefore $\sin(x) = 9.7 \sin(115 \text{ degrees})/18.606... = 0.472...$.*Why: Rearrange the sine-rule equation for $\sin(x)$.***Method** $x = \sin^{-1}(0.472...) = 28.2 \text{ degrees to 3 significant figures.}$ *Why: Take the inverse sine in degree mode and round once at the end.***Final answer****angle:** $x = 28.2 \text{ degrees}$

Original question 13

4MA1-1H Q19 Q19

ORIGINAL QUESTION**5 marks****19**Diagram **NOT**
accurately drawn

OAB is a sector **S** of a circle with centre O and radius $(r + 7)$ metres.
Angle $AOB = 45^\circ$

A circle **C** has radius $(r - 2)$ metres.

The area of sector **S** is twice the area of circle **C**

Find the value of r

Show your working clearly.

Question 19 continued. $r = \dots\dots\dots$

Worked solution 13

Chapter: Circles, Arcs & Sectors

MODULAR UNIT 1

5 marks

Q19 – Chapter: Circles, Arcs & Sectors

Family: Resolve compound and shaded circle regions • Variant: Solve an algebraic circle-measure constraint

Method

The sector area is $(45/360)\pi(r + 7)^2$.

Why: A 45-degree sector is $45/360$ of its full circle.

Method

The condition gives $(45/360)\pi(r + 7)^2 = 2\pi(r - 2)^2$.

Why: The sector area is twice the area of circle C.

Method

After cancelling π and simplifying, $(r + 7)^2 = 16(r - 2)^2$, equivalently $5r^2 - 26r + 5 = 0$.

Why: Expand or rearrange the exact area equation.

Method

$5r^2 - 26r + 5 = (5r - 1)(r - 5)$, so $r = 1/5$ or $r = 5$.

Why: Factorise and apply the zero-product rule.

Method

A circle of radius $r - 2$ requires $r > 2$, so $r = 1/5$ is invalid and $r = 5$.

Why: Use the physical positivity of the stated radius.

Final answer

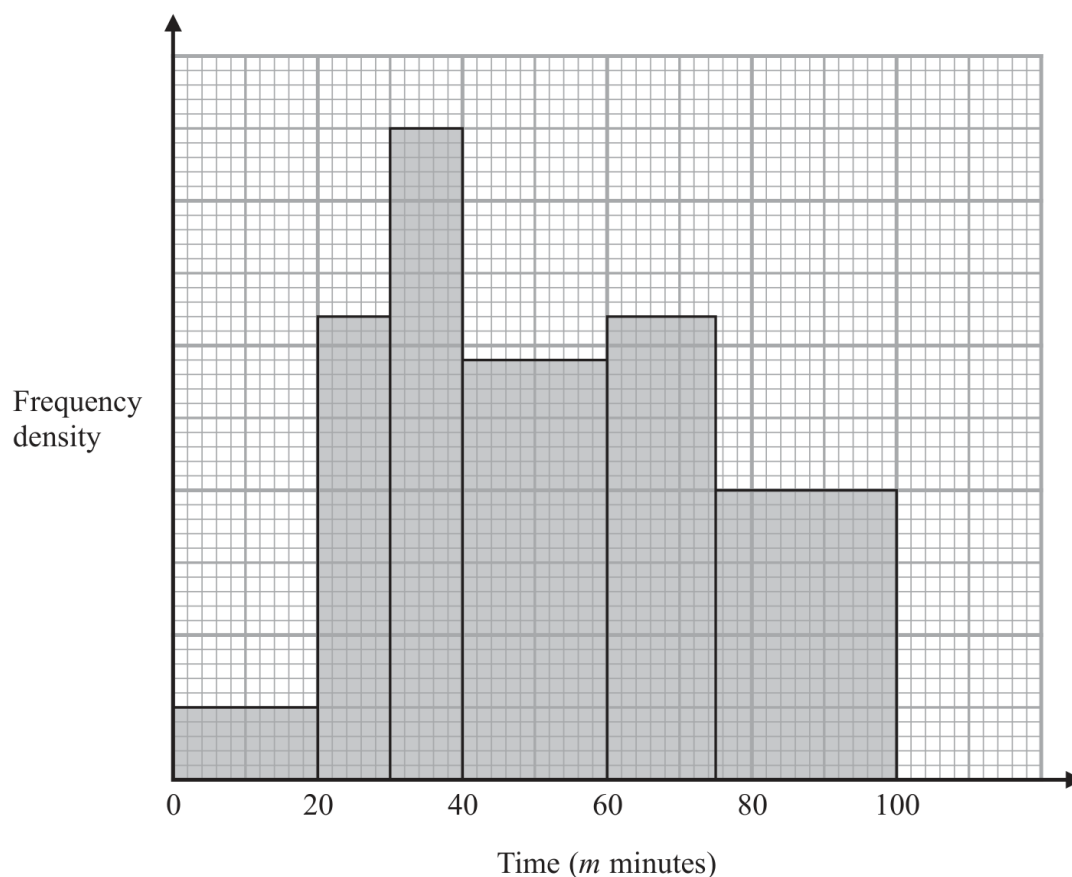
r: $r = 5$

Original question 14

4MA1-1H Q21 Q21

ORIGINAL QUESTION**3 marks**

- 21 The histogram shows information about the total time, m minutes, taken by each child in a school to walk to school every day for one week.



There are no children for whom $m > 100$

There are 10 children for whom $m \leq 20$

Work out an estimate for the number of children for whom $50 < m \leq 80$

Worked solution 14

Chapter: Histograms

MODULAR UNIT 1

3 marks

Q21 – Chapter: Histograms

Family: Interpret histogram areas • Variant: Estimate a partial-class frequency

Method

For $0 < m \leq 20$, frequency density = $10/20 = 0.5$; use this bar to calibrate the vertical scale.

Why: Histogram area equals frequency, and density equals frequency divided by class width.

Method

From the calibrated grid, the densities over 50-60, 60-75, and 75-80 are approximately 2.9, 3.2, and 2.0.

Why: Split the requested interval wherever a histogram class boundary occurs.

Method

Estimated frequency = $10(2.9) + 15(3.2) + 5(2.0) = 29 + 48 + 10 = 87$ children.

Why: Add the areas of the three relevant histogram pieces.

Final answer

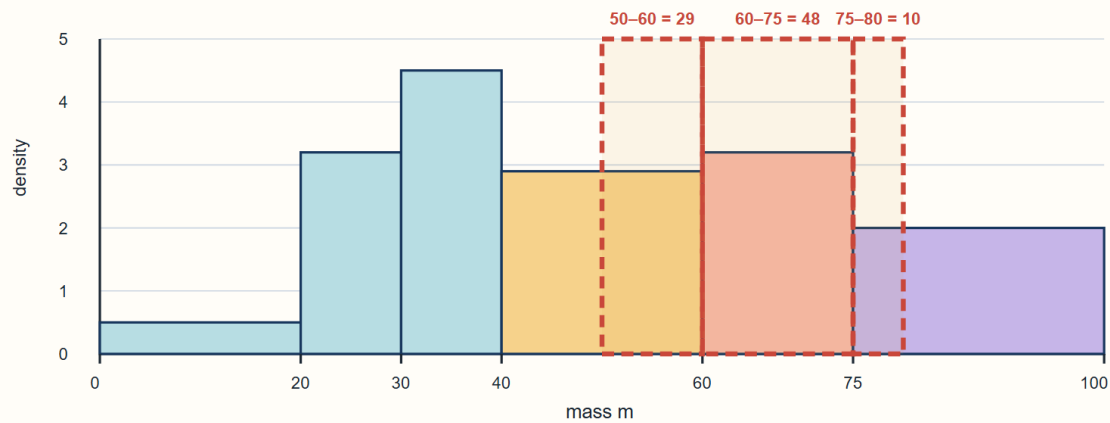
estimated children: 87

Completed diagram 14

Chapter: Histograms

MODULAR UNIT 1

3 marks

Q21(a) Annotated histogram readingDensity $\times$ class width gives frequency; split 50–80 at 60 and 75Target estimate: $10 \times 2.9 + 15 \times 3.2 + 5 \times 2.0 = 87$ Check: frequency = density $\times$ width

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Original question 15

4MA1-1H Q24 Q24

ORIGINAL QUESTION**4 marks**

24 Express each of a , b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$$a = \dots\dots\dots$$

$$b = \dots\dots\dots$$

$$c = \dots\dots\dots$$

Worked solution 15

Chapter: Completing the Square

MODULAR UNIT 1

4 marks

Q24 – Chapter: Completing the Square

Family: Complete the square for a non-monic quadratic • Variant: Negative leading coefficient or maximum form

Method

$$q + 12x - qx^2 = -q[x^2 - (12/q)x] + q.$$

Why: Factor $-q$ from the quadratic and linear x -terms.

Method

$$x^2 - (12/q)x = (x - 6/q)^2 - 36/q^2.$$

Why: Complete the square by halving the coefficient of x .

Method

$$\text{Therefore } q + 12x - qx^2 = -q(x - 6/q)^2 + 36/q + q.$$

Why: Substitute the completed-square identity and distribute $-q$.

Method

$$\text{Matching } a - b(x - c)^2 \text{ gives } a = q + 36/q, b = q, \text{ and } c = 6/q.$$

Why: Compare the constant, outside coefficient, and horizontal-shift terms.

Final answer

$$\mathbf{a: } a = q + 36/q$$

Final answer

$$\mathbf{b: } b = q$$

Final answer

$$\mathbf{c: } c = 6/q$$

Original question 16

4MA1-2H Q3 Q3

ORIGINAL QUESTION**3 marks**

- 3 An aeroplane travelled from New York City to Los Angeles.

The aeroplane travelled a distance of 3980 kilometres in 5 hours 24 minutes.

Work out the average speed of the aeroplane.

Give your answer in kilometres per hour correct to the nearest whole number.

..... kilometres per hour

Worked solution 16

Chapter: Standard & Compound Units

MODULAR UNIT 1

3 marks

Q3 – Chapter: Standard & Compound Units

Family: Use speed, distance and time • Variant: Find average speed for one stage

Method $5 \text{ hours } 24 \text{ minutes} = 5 + 24/60 = 5.4 \text{ hours.}$ *Why: Convert minutes to hours before using kilometres per hour.***Method** $\text{Average speed} = 3980/5.4 = 737.037... \text{ kilometres per hour.}$ *Why: Average speed is distance divided by time.***Method**

To the nearest whole number, the speed is 737 kilometres per hour.

*Why: Round the unrounded quotient once at the end.***Final answer****speed:** 737 km/h

Original question 17

4MA1-2H Q4 Q4

ORIGINAL QUESTION**3 marks**

4 Show that $5\frac{1}{3} - 2\frac{6}{7} = 2\frac{10}{21}$

Worked solution 17

Chapter: Fractions

MODULAR UNIT 1

3 marks

Q4 – Chapter: Fractions

Family: Add or subtract fractions and mixed numbers • Variant: Subtract fractions or mixed numbers

Method

$$5 \frac{1}{3} - 2 \frac{6}{7} = \frac{16}{3} - \frac{20}{7}.$$

*Why: Convert both mixed numbers to improper fractions.***Method**

$$\frac{16}{3} - \frac{20}{7} = \frac{112}{21} - \frac{60}{21} = \frac{52}{21}.$$

*Why: Use a common denominator of 21.***Method**

$$\frac{52}{21} = 2 \frac{10}{21}, \text{ which is the required result.}$$

*Why: Convert the improper fraction back to a mixed number.***Final answer****shown value:** $2 \frac{10}{21}$

Original question 18

4MA1-2H Q5 Q5

ORIGINAL QUESTION**4 marks**

- 5 The diagram shows an 8-sided shape $ABCDEFGH$.

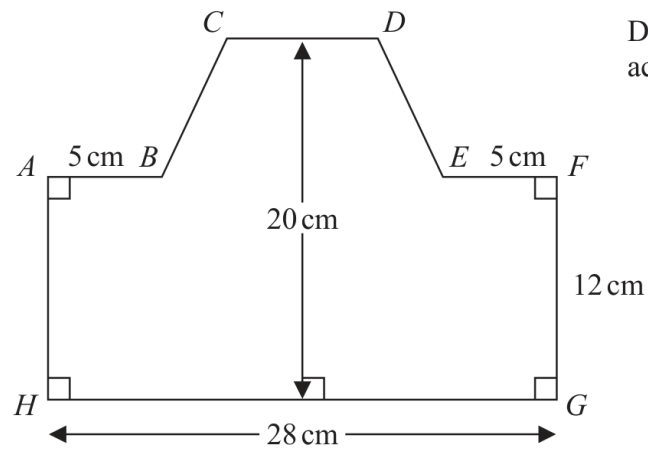


Diagram **NOT**
accurately drawn

$$HG = 28 \text{ cm} \quad FG = 12 \text{ cm} \quad AB = EF = 5 \text{ cm}$$

The height of the shape is 20 cm

CD is parallel to HG

The area of shape $ABCDEFGH$ is 434 cm^2

Find the length of CD .

..... cm

Worked solution 18

Chapter: Area & Perimeter

MODULAR UNIT 1

4 marks

Q5 – Chapter: Area & Perimeter

Family: Use parallelogram and trapezium area • Variant: Recover a missing trapezium or parallelogram dimension

Method

The lower rectangle has area $28 \times 12 = 336 \text{ cm}^2$.

Why: Its width and height are given by HG and FG.

Method

The top section has height $20 - 12 = 8 \text{ cm}$ and lower parallel side $28 - 5 - 5 = 18 \text{ cm}$.

Why: Remove the two 5 cm horizontal offsets.

Method

If $CD = x$, then $336 + \frac{1}{2}(18 + x) \times 8 = 434$.

Why: Add the rectangle area to the trapezium area.

Method

$4(18 + x) = 98$, so $x = 6.5$ and $CD = 6.5 \text{ cm}$.

Why: Solve the resulting linear equation.

Final answer

CD: $CD = 6.5 \text{ cm}$

Original question 19

4MA1-2H Q6 Q6

ORIGINAL QUESTION**3 marks**

- 6 The diagram shows triangle PQR .

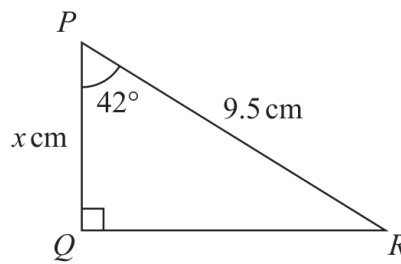


Diagram **NOT**
accurately drawn

Work out the value of x
Give your answer correct to one decimal place.

$x =$

Worked solution 19

Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

MODULAR UNIT 1

3 marks

Q6 – Chapter: Right-Angled Triangles - Pythagoras & Trigonometry

Family: Use right-triangle trigonometry for lengths • Variant: Find adjacent side with cosine or tangent

MethodPR is the hypotenuse and PQ = x is adjacent to 42 degrees, so $\cos 42 \text{ degrees} = x/9.5$.*Why: Use cosine for adjacent over hypotenuse.***Method** $x = 9.5 \cos 42 \text{ degrees} = 7.059... .$ *Why: Rearrange and evaluate in degree mode.***Method**Therefore $x = 7.1$ to one decimal place.*Why: Round once at the requested accuracy.***Final answer****x:** $x = 7.1$

Original question 20

4MA1-2H Q7 Q7

ORIGINAL QUESTION**3 marks**

7 Change a speed of 81 kilometres per hour to a speed in metres per second.

..... metres per second

Worked solution 20

Chapter: Standard & Compound Units

MODULAR UNIT 1

3 marks

Q7 – Chapter: Standard & Compound Units

Family: Use speed, distance and time • Variant: Convert compound speed units

Method $81 \text{ km/h} = 81 \times 1000 \text{ metres per hour.}$ *Why: One kilometre is 1000 metres.***Method** $81 \times 1000 / (60 \times 60) = 22.5 \text{ metres per second.}$ *Why: One hour is 3600 seconds.***Final answer****speed: 22.5 m/s**

Original question 21

4MA1-2H Q11 Q11(i), Q11(ii)

ORIGINAL QUESTION**3 marks****11** (i) Factorise $x^2 + 5x - 24$

(2)(ii) Hence, solve $x^2 + 5x - 24 = 0$

(1)

Worked solution 21

Chapter: Factorising; Solving Quadratic Equations

MODULAR UNIT 1

3 marks

Q11(i) – Chapter: Factorising

Family: Factorise a monic quadratic • Variant: Negative constant monic quadratic

Method

The required numbers have product -24 and sum 5, so they are 8 and -3.

*Why: Match the constant term and middle coefficient.***Method**Therefore $x^2 + 5x - 24 = (x - 3)(x + 8)$.*Why: Write the monic quadratic as two linear factors.***Final answer****factorisation:** $(x - 3)(x + 8)$ **Q11(ii) – Chapter: Solving Quadratic Equations**

Family: Solve a quadratic by factorisation • Variant: Monic quadratic roots by factors

MethodFrom $(x - 3)(x + 8) = 0$, $x = 3$ or $x = -8$.*Why: Use the zero-product rule on the factors from part (i).***Final answer****roots:** $x = 3$ or $x = -8$

Original question 22

4MA1-2H Q14 Q14(a)

ORIGINAL QUESTION**3 marks**

- 14 (a)** Expand and simplify $(5 - x)(2x + 3)(x + 4)$
Show your working clearly.

(3)

Worked solution 22

Chapter: Expanding Brackets

MODULAR UNIT 1

3 marks

Q14(a) – Chapter: Expanding Brackets

Family: Expand three or more linear factors • Variant: Three linear brackets

Method

$$(5 - x)(2x + 3) = -2x^2 + 7x + 15.$$

*Why: Expand and collect the first pair of factors.***Method**

$$(-2x^2 + 7x + 15)(x + 4) = -2x^3 - 8x^2 + 7x^2 + 28x + 15x + 60.$$

*Why: Distribute every term across the third factor.***Method**The simplified result is $-2x^3 - x^2 + 43x + 60$.*Why: Collect like powers of x .***Final answer**expansion: $-2x^3 - x^2 + 43x + 60$

Original question 23

4MA1-2H Q15 Q15(a)

ORIGINAL QUESTION**4 marks**

15 (a) Solve $\frac{4x + 5}{3} - \frac{3 - 2x}{2} = 13$

Show clear algebraic working.

$x = \dots\dots\dots$
(4)

Worked solution 23

Chapter: Solving Linear Equations

MODULAR UNIT 1

4 marks

Q15(a) – Chapter: Solving Linear Equations

Family: Clear fractional linear equations • Variant: Several fractional linear terms

MethodMultiply by 6: $2(4x + 5) - 3(3 - 2x) = 78$.*Why: Use the lowest common denominator to clear both fractions.***Method** $8x + 10 - 9 + 6x = 78$.*Why: Expand both brackets, preserving the subtraction.***Method** $14x = 77$.*Why: Collect x -terms and constants.***Method** $x = 77/14 = 11/2 = 5.5$.*Why: Divide by 14 and simplify.***Final answer****x:** $x = 5.5$

Original question 24

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

ORIGINAL QUESTION**8 marks**

16 100 farmers are asked if they have goats (G), sheep (S) or chickens (C) on their farms.

Of these farmers

31 have sheep

53 have chickens

6 have goats, sheep and chickens

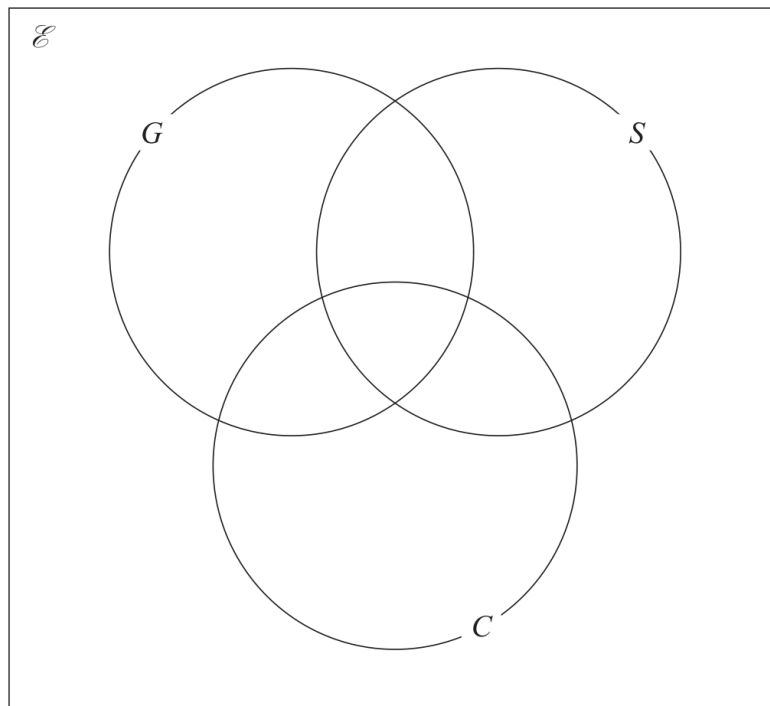
11 have sheep and goats

17 have sheep and chickens

18 have goats and chickens

20 do not have any goats, sheep or chickens

(a) Using this information, complete the Venn diagram to show the number of farmers in each appropriate subset.

**(3)**

(b) Find

Original question 24 (continued)

4MA1-2H Q16 Q16(a), Q16(b)(i), Q16(b)(ii), Q16(b)(iii), Q16(c)

ORIGINAL QUESTION**8 marks**

(i) $n(G)$

.....
(1)

(ii) $n([G \cup S]')$

.....
(1)

(iii) $n(G' \cap C)$

.....
(1)

One of the farmers who has chickens is chosen at random.

(c) Find the probability that this farmer also has goats.

.....
(2)

Worked solution 24

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1

8 marks

Q16(a) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Complete a three-set Venn diagram

Method

Put 6 in the triple intersection; then GS-only = $11 - 6 = 5$, SC-only = $17 - 6 = 11$, and GC-only = $18 - 6 = 12$.

Why: The pair totals include the triple intersection.

Method

S-only = $31 - (5 + 11 + 6) = 9$ and C-only = $53 - (11 + 12 + 6) = 24$.

Why: Subtract every already-filled sheep or chicken region.

Method

There are 80 farmers inside the three sets, so G-only = $80 - (9 + 24 + 5 + 11 + 12 + 6) = 13$; put 20 outside.

Why: Use the universal total and the 20 with none.

Final answer

Venn regions: G only 13; S only 9; C only 24; GS only 5; GC only 12; SC only 11; GSC 6; outside 20

Q16(b)(i) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$n(G) = 13 + 5 + 12 + 6 = 36$.

Why: Add every region inside G.

Final answer

$n(G)$: 36

Q16(b)(ii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

$(G \cup S)'$ contains C-only and the outside region, so $n((G \cup S)') = 24 + 20 = 44$.

Why: The complement excludes every point in G or S.

Final answer

count: 44

Q16(b)(iii) – Chapter: Probability Diagrams - Venn & Tree Diagrams

Family: Represent and read events with Venn diagrams • Variant: Read regions and probabilities from a Venn diagram

Method

G' intersection C contains C-only and SC-only, so $n(G' \text{ intersection } C) = 24 + 11 = 35$.

Why: Include chicken regions that are outside G.

Final answer

count: 35

Q16(c) – Chapter: Combined & Conditional Probability

Family: Use conditional probability and restricted sample spaces • Variant: Calculate a direct given-that probability

Method

The chosen farmer comes from the 53 farmers with chickens.

Why: This is the conditional sample space.

Method

Of these, 18 also have goats, so the probability is $18/53$.

Why: Use the G-and-C count over the C count.

Final answer

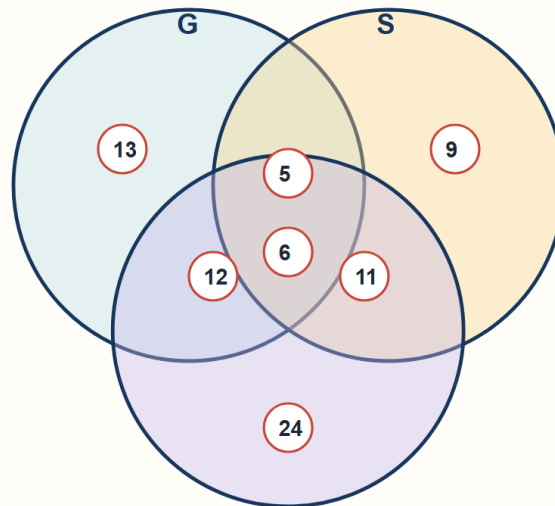
probability: $18/53$

Completed diagram 24

Chapter: Probability Diagrams - Venn & Tree Diagrams; Combined & Conditional Probability

MODULAR UNIT 1**8 marks****Q16(a) Completed three-set Venn diagram**

Place every region, including the outside count



outside

20

Inside all three = 6

Inside at least one = 80

Total farmers = 100

 $80 + 20 = 100 \checkmark$

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Original question 25

4MA1-2H Q18 Q18

ORIGINAL QUESTION**3 marks**

18 $X = \frac{2a - b}{f}$

$a = 7.5$ correct to 1 decimal place.

$b = 3.42$ correct to 2 decimal places.

$f = 2$ correct to the nearest whole number.

Work out the upper bound of the value of X

Show your working clearly.

Worked solution 25

Chapter: Rounding, Estimation & Bounds

MODULAR UNIT 1

3 marks

Q18 – Chapter: Rounding, Estimation & Bounds

Family: Optimise a compound expression using bounds • Variant: Algebraic expression upper or lower bound

Method

The relevant bounds are $a < 7.55$, $b \geq 3.415$, and $f \geq 1.5$.

Why: Use the stated rounding accuracies.

Method

Because the numerator and denominator are positive, the upper bound is $(2 \times 7.55 - 3.415)/1.5$.

Why: Maximise the numerator and minimise the positive denominator.

Method

The upper bound is $11.685/1.5 = 7.79$.

Why: Evaluate the extremal expression exactly to the terminating decimal.

Final answer

upper bound: 7.79

Original question 26

4MA1-2H Q19 Q19

ORIGINAL QUESTION**3 marks**

19 $a = \frac{14}{3x - 7} \quad x = \frac{7}{4y - 3}$

Express a in the form $\frac{py + q}{ry + s}$ where p, q, r and s are integers.

Give your answer in its simplest form.

$a = \dots\dots\dots$

Worked solution 26

Chapter: Algebraic Fractions

MODULAR UNIT 1

3 marks

Q19 – Chapter: Algebraic Fractions

Family: Substitute or transform a rational expression • Variant: Substitute a variable relation into a fraction

MethodSubstitute $x = 7/(4y - 3)$: $a = 14/[21/(4y - 3) - 7]$.*Why: Replace x in the definition of a .***Method** $a = 14(4y - 3)/[21 - 7(4y - 3)] = 14(4y - 3)/(42 - 28y)$.*Why: Combine the denominator into one fraction.***Method**Cancel the common factor 14: $a = (4y - 3)/(3 - 2y)$.*Why: Factor and simplify completely.***Final answer****a:** $a = (4y - 3)/(3 - 2y)$

Original question 27

4MA1-2H Q20 Q20

ORIGINAL QUESTION**4 marks**

- 20 The diagram shows four identical circles drawn inside a square.

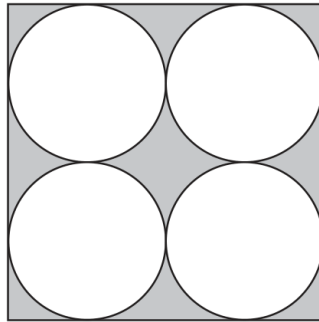


Diagram **NOT**
accurately drawn

Each circle touches two other circles and two sides of the square.

The region inside the square that is outside the circles, shown shaded in the diagram, has a total area of 40 cm^2

Work out the perimeter of the square.

Give your answer correct to 3 significant figures.

..... cm

Worked solution 27

Chapter: Circles, Arcs & Sectors

MODULAR UNIT 1

4 marks

Q20 – Chapter: Circles, Arcs & Sectors

Family: Resolve compound and shaded circle regions • Variant: Find annular, overlapping or multi-circle area

Method

Let each circle have radius r ; the square side is $4r$.

Why: Two diameters fit along each side.

Method

The shaded area is $(4r)^2 - 4\pi r^2 = 40$, so $r^2 = 10/(4 - \pi)$.

Why: Subtract the four circle areas from the square area.

Method

The square perimeter is $16r = 16\sqrt{10/(4 - \pi)}$.

Why: Perimeter is four times the side $4r$.

Method

This is 54.608... cm, so the perimeter is 54.6 cm to three significant figures.

Why: Round the final value once.

Final answer

perimeter: 54.6 cm

Original question 28

4MA1-2H Q22 Q22

ORIGINAL QUESTION**7 marks**

- 22** $ABCD$ is a kite, with diagonals AC and BD , drawn on a centimetre square grid, with a scale of 1 cm for 1 unit on each axis.

A is the point with coordinates $(-3, 4)$

The diagonals of the kite intersect at the point M with coordinates $(0, 2)$

Given that $AB = AD = 6.5$ cm and the x coordinate of B is positive,

find the coordinates of the points B and D .

(.....,)

(.....,)

Worked solution 28

Chapter: Coordinate Geometry

MODULAR UNIT 1

7 marks

Q22 – Chapter: Coordinate Geometry

Family: Determine coordinates from geometric information • Variant: Kite or symmetric vertex coordinates

Method

The gradient of AM is $(2 - 4) / (0 - (-3)) = -2/3$.

Why: Use the two given coordinates.

Method

The kite diagonals are perpendicular, so BD has gradient $3/2$ and equation $y = (3/2)x + 2$.

Why: Use the negative reciprocal and point M.

Method

For B or D, $(x + 3)^2 + (y - 4)^2 = 6.5^2$.

Why: Each point is 6.5 units from A.

Method

Substitute $y = (3/2)x + 2$: $(x + 3)^2 + ((3/2)x - 2)^2 = 42.25$.

Why: Combine the line and distance conditions.

Method

Expanding gives $3.25x^2 + 13 = 42.25$.

Why: Expand and collect; the linear x terms cancel.

Method

Thus $x^2 = 9$, so $x = 3$ or $x = -3$; the corresponding y -values are 6.5 and -2.5.

Why: Solve the quadratic and substitute into the line.

Method

Since B has positive x -coordinate, $B = (3, 6.5)$ and $D = (-3, -2.5)$.

Why: Use the final labelling condition.

Final answer

B: $B = (3, 6.5)$

Final answer

D: $D = (-3, -2.5)$

Original question 29

4MA1-2H Q24 Q24

ORIGINAL QUESTION**3 marks****24**

$$\frac{18 \times (\sqrt{27})^{4n+6}}{6 \times 9^{2n+8}} = 3^x$$

Express x in terms of n

Show your working clearly and simplify your expression.

 $x = \dots\dots\dots$

Worked solution 29

Chapter: Algebraic Roots & Indices

MODULAR UNIT 1

3 marks

Q24 – Chapter: Algebraic Roots & Indices

Family: Solve an index relation by a common base • Variant: Express one exponent variable in terms of others

Method

Use $18 = 2 \times 3^2$, $\sqrt{27} = 3^{(3/2)}$, $6 = 2 \times 3$, and $9 = 3^2$.

Why: Rewrite every factor using base 3 where possible.

Method

The left side becomes $[2 \times 3^2 \times 3^{(6n+9)}] / [2 \times 3 \times 3^{(4n+16)}]$.

Why: Multiply the nested exponents and collect the fixed factors.

Method

This simplifies to $3^{(6n+11-4n-17)} = 3^{(2n-6)}$, so $x = 2n - 6$.

Why: Subtract denominator exponents and equate equal bases.

Final answer

x: $x = 2n - 6$